POSEBUSTERS BENCHMARK · LIGAND-CONDITIONED RE-ANALYSIS · INDEPENDENT WORK

Which molecules break docking predictions

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The published benchmark records which docking *method* fails which physical-validity check. It carries no chemistry about the *molecule* being docked. This joins RDKit descriptors from the crystal ligands onto every pose and asks what the failures have in common — with eligibility gating, per-method strata, cluster-bootstrap intervals, FDR control, an external validation and a held-out replication.

7,695 POSE ROWS JOINED	**41/126** STRATA ESTIMABLE	**95/287** EFFECTS SURVIVE FDR
99.8% WORST-CASE VALIDATION	**8/14** REPLICATE HELD-OUT	**37** TESTS · 20 COMMITS

THE BENCHMARK 01

Accuracy and validity are different questions

Each of 7 docking methods predicted a pose for all 428 complexes. Two things are then asked of every pose: is it in the right *place* (RMSD ≤ 2 Å against the crystal structure), and is it a *molecule that could exist* (18 physical-validity checks — bond lengths and angles within distance-geometry bounds, aromatic rings planar, no internal or protein clashes, stereochemistry preserved, strain energy not absurd).

A pose can pass one and fail the other, and that gap is the subject of this report.

METHOD	CLASS	RMSD ≤ 2 Å	VALID	BOTH	OF ACCURATE POSES, INVALID
vina	classical	52.3%	97.4%	51.2%	2.2%
gold	classical	51.2%	93.9%	47.7%	6.8%
diffdock	deep learning	37.9%	24.1%	13.6%	64.2%
unimol	deep learning	22.9%	5.6%	1.9%	91.8%
deepdock	deep learning	17.8%	8.2%	4.7%	73.7%
tankbind	deep learning	15.0%	4.7%	2.6%	82.8%
equibind	deep learning	2.6%	0.9%	0.2%	90.9%

Classical search methods produce accurate poses about half the time and nearly all of those are physically valid. The deep-learning methods' accurate poses are mostly invalid. These figures reproduce the paper and nothing below revises them.

VALIDATION 02

The published columns mean what we think they mean

Every number in this report reads the paper's published check results rather than recomputing them — so the whole analysis rests on an assumption about what each column means. That assumption was tested by re-running PoseBusters itself on all 513 crystal ligands against their own receptors and comparing to the paper's own reference rows.

Worst-case agreement across 17 mapped checks is `99.81%`; 15 of 17 agree on all 513 structures. The two disagreements are single structures on different checks — `7V3S` on ring flatness and `7T0U` on strain energy, the latter one of the paper's own two acknowledged borderline failures.

CHECK	COMPARED	AGREE	AGREEMENT
aromatic ring flatness passes	513	512	99.81%
energy ratio within threshold	513	512	99.81%
molecular formula preserved	513	513	100.00%
molecular bonds preserved	513	513	100.00%
sp3 stereochemistry preserved	513	513	100.00%
double bond stereochemistry preserved	513	513	100.00%

Two thirds of the grid cannot be measured at all

There are 126 method × check combinations. Only 41 carry enough failures and enough distinct ligands to estimate an effect. The rest are not null results — they are unmeasured, and the distinction matters when reading anything below.

OUTCOME	STRATA	SHARE
Check never failed for that method	59	47%
Estimated	41	33%
Fewer than 15 failures	25	20%
Fewer than 30 distinct ligands	1	1%

Across those 41 strata every descriptor was tested against every check, giving 287 estimates, of which 95 survive Benjamini–Hochberg correction at $\alpha = 0.05$. Intervals are cluster-bootstrapped on ligand, because each ligand contributes one row per method and treating those as independent would report intervals about $\sqrt{5}$ too narrow.

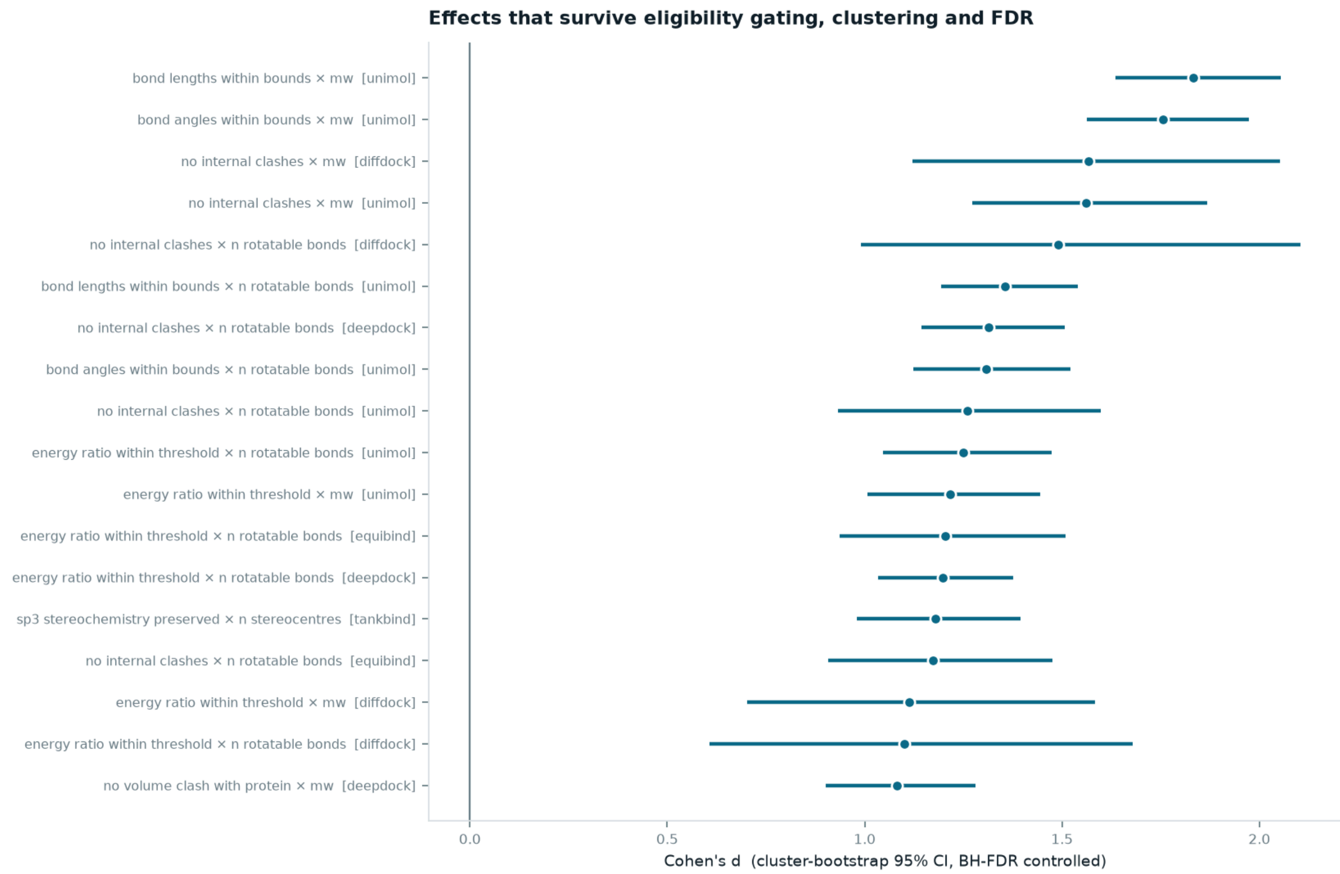

FIG 1 The 18 largest surviving marginal effects, with cluster-bootstrap intervals. Read as associations that survive gating, clustering and multiplicity correction — not as a ranking of causes. Note how `mw` and `n rotatable bonds` appear on the same check for the same method: that is one signal counted twice.

METHOD	CHECK	DESCRIPTOR	D	95% CI	FAILURES	LIGANDS
unimol	bond lengths within bounds	mw	1.83	[1.64, 2.05]	269	427
unimol	bond angles within bounds	mw	1.76	[1.56, 1.97]	284	427
diffdock	no internal clashes	mw	1.57	[1.12, 2.05]	25	426
unimol	no internal clashes	mw	1.56	[1.27, 1.87]	66	427
diffdock	no internal clashes	n rotatable bonds	1.49	[0.99, 2.10]	25	426
unimol	bond lengths within bounds	n rotatable bonds	1.36	[1.19, 1.54]	269	427
deepdock	no internal clashes	n rotatable bonds	1.31	[1.14, 1.51]	205	427
unimol	bond angles within bounds	n rotatable bonds	1.31	[1.12, 1.52]	284	427
unimol	no internal clashes	n rotatable bonds	1.26	[0.93, 1.60]	66	427
unimol	energy ratio within threshold	n rotatable bonds	1.25	[1.05, 1.47]	165	427
unimol	energy ratio within threshold	mw	1.22	[1.01, 1.45]	165	427
equibind	energy ratio within threshold	n rotatable bonds	1.20	[0.94, 1.51]	92	426

Flexibility or size? Not resolvable for the method it was asked about

The obvious hypothesis is that torsional freedom, not molecular size, drives strain and clash failures. Testing it requires holding one fixed while varying the other — but `mw` and `n_rotatable_bonds` correlate at `ρ ≈ 0.73–0.74 within every method's ligand set`, so a stratified table cannot separate them. A multivariate model can try. One clustered logistic regression per method, both descriptors and five others entered together, on the strain check:

METHOD	P (MW)	P (ROTATABLE BONDS)	VERDICT
diffdock	0.941	0.108	Neither — not resolvable
equibind	0.956 (negative)	0.000	Flexibility only
deepdock	0.020 (negative)	0.000	Both carry independent signal
tankbind	0.044	0.001	Both carry independent signal
unimol	0.010	0.002	Both carry independent signal
gold	0.471 (negative)	0.395	Neither — not resolvable
vina	—	—	No model fit (too few failures)

DiffDock is the method the question was mainly about, and for DiffDock neither descriptor clears significance once the other is held fixed — despite both individually surviving FDR in the marginal grid. EquiBind gives the one clean flexibility-only result, and it is the weakest method in the benchmark. Three methods show both descriptors carrying independent signal; DeepDock's size coefficient is negative, the opposite sign from a naive "bigger molecules fail more" story.

The honest conclusion is that this dataset cannot adjudicate size versus flexibility for the method that matters. That is a limitation, not a finding.

Crystal contacts do not inflate protein-clash failure

The benchmark's authors dropped 120 of 428 complexes for their journal version because the ligand touches a crystallographic symmetry mate, on the reasoning that such contacts inflate protein-clash failures. That list was never published, so the flag was recomputed here from first principles — expand each deposited structure by its spacegroup, measure the closest approach to any symmetry image of the protein. That gives 104 complexes with a genuine protein contact, or 119 counting solvent and cryoprotectant, bracketing the paper's 120.

With intervals on the contact-minus-no-contact difference, the prediction is **not supported**: one method of seven shows a significant difference, and it runs in the *opposite* direction.

METHOD	CONTACT	NO CONTACT	DIFFERENCE (PP)	95% CI	EXCLUDES ZERO
unimol	68.9% (103)	86.1% (323)	−17.1	[−27.0, −7.7]	yes
diffdock	61.2% (103)	70.5% (322)	−9.3	[−19.6, +1.8]	no
tankbind	77.9% (104)	85.1% (323)	−7.3	[−16.2, +1.6]	no
equibind	96.1% (103)	98.4% (322)	−2.3	[−6.6, +1.2]	no
gold	2.9% (102)	5.0% (322)	−2.0	[−5.9, +2.4]	no
vina	0.0% (104)	0.6% (323)	−0.6	[−1.5, +0.0]	no
deepdock	91.3% (103)	85.1% (323)	+6.1	[−0.9, +12.5]	no

Only Uni-Mol's interval excludes zero, and its clash failure is *lower* among contact-bearing complexes. The other six are indistinguishable from zero and none of their point estimates should be read as a direction — Vina's entire contribution is 2 failing poses out of 323, Gold's is 16 of 322 against 3 of 102. At those counts a direction is noise from a handful of poses.

RETRACTED

The cofactor-chemistry finding is withdrawn. An earlier version of this analysis reported that small, non-aromatic ligands preferentially clash with cofactors — `d = −0.29`, 95% CI [-0.45, -0.12], $p = 0.0005$, pooled across all 2,987 eligible rows.

That pooling was the error. Over half those rows are complexes with *no cofactor at all*, where the check passes almost automatically, so the contrast mostly measured whether the receptor has a cofactor rather than any property of the ligand. Restricted to the 1,304 rows where a cofactor is actually present — the only population in which the check can be informative — the effect collapses to `d = −0.03`, 95% CI [-0.21, 0.16], $p = 0.765$. The finding is deleted, not restated more cautiously.

REPLICATION 06

Held-out Astex Diverse Set

Every effect above was discovered and estimated on the same 428 complexes. The 85 Astex complexes were never touched during the analysis, so re-estimating a shortlist there is an independent check. Of 14 checkable triples, **8 replicate with statistical support** — both intervals excluding zero. 13 of 14 agree in sign, but sign agreement between two estimates that both straddle zero carries no information, so 8/14 is the number to cite. 2 rows rest on fewer than 15 failures and are flagged.

METHOD	CHECK	DESCRIPTOR	D (BENCHMARK)	D (ASTEX)	REPLICATED	THIN
diffdock	no clashes with protein	n rotatable bonds	0.60	0.30	no	
unimol	energy ratio within threshold	n rotatable bonds	1.25	0.96	yes	
unimol	no internal clashes	n rotatable bonds	1.26	0.87	yes	thin
unimol	sp3 stereochemistry preserved	n stereocentres	0.84	0.25	no	
unimol	no clashes with protein	n rotatable bonds	0.54	0.61	yes	
deepdock	energy ratio within threshold	n rotatable bonds	1.20	1.38	yes	
deepdock	no internal clashes	n rotatable bonds	1.31	1.14	yes	
deepdock	sp3 stereochemistry preserved	n stereocentres	0.05	0.03	no	
deepdock	no clashes with protein	n rotatable bonds	0.92	0.65	yes	
tankbind	energy ratio within threshold	n rotatable bonds	0.97	1.03	yes	
tankbind	no internal clashes	n rotatable bonds	0.86	0.47	yes	
tankbind	sp3 stereochemistry preserved	n stereocentres	1.18	0.17	no	
tankbind	no clashes with protein	n rotatable bonds	0.37	0.22	no	
gold	no clashes with protein	n rotatable bonds	0.15	-1.00	no	thin

The strain and internal-clash effects against rotatable bonds are the most robust of the shortlist. The stereochemistry effect and the protein-clash effect are significant on the benchmark but do not replicate with support on Astex for most methods.

Validity across the whole RMSD distribution

Reporting "of accurate poses, the share invalid" conditions on the accuracy outcome, which shares an upstream cause with validity — how well the method handled that ligand. Conditioning on such a variable can manufacture association. Banding raw RMSD instead needs no conditioning and uses every pose, not only those clearing an arbitrary 2 Å cutoff.

METHOD	(0.0, 1.0]	(1.0, 2.0]	(2.0, 3.0]	(3.0, 5.0]	(5.0, 10.0]	(10.0, INF]
deepdock	55.0%	16.1%	12.5%	6.8%	1.1%	0.0%
diffdock	61.3%	20.0%	20.0%	17.8%	18.6%	12.7%
gold	95.2%	88.9%	95.0%	90.7%	97.6%	100.0%
tankbind	36.4%	13.2%	3.0%	1.8%	0.0%	6.9%
unimol	33.3%	2.5%	6.2%	5.2%	4.2%	0.0%
vina	100.0%	93.5%	97.6%	96.2%	96.2%	100.0%

The deep-learning methods' validity is already poor at sub-Ångström accuracy and does not recover at larger RMSD. That is a stronger statement than "poses that barely clear 2 Å are usually invalid" — it holds across the whole distribution rather than at one threshold.

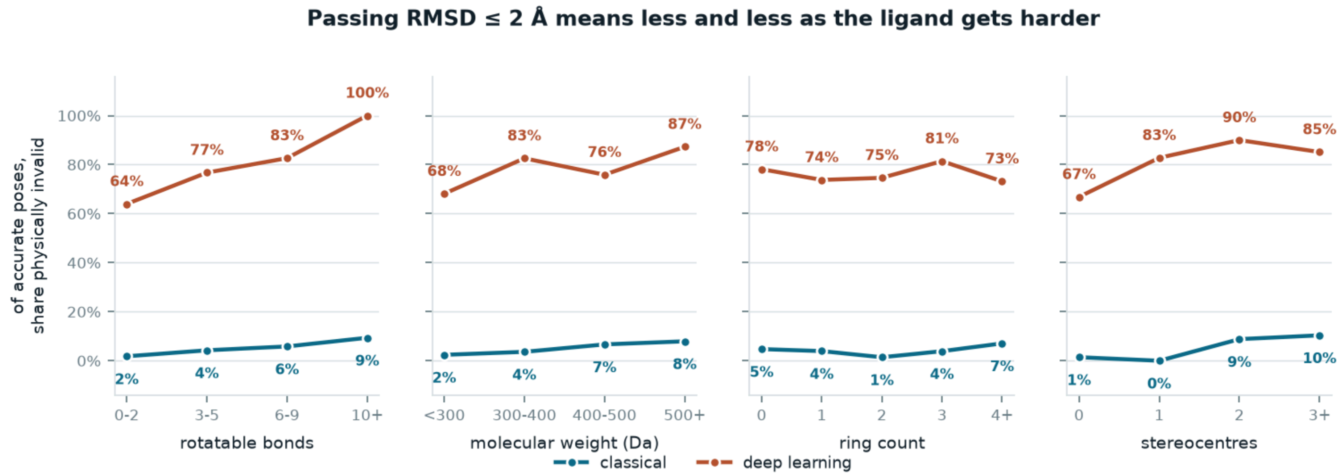

FIG 2 The conditioned view, kept as secondary. Monotone against rotatable bonds in both method classes; close to flat against ring count. The partition-level trend is specific to flexibility among these four — but see section 04: the descriptors are correlated throughout, so this alone does not establish cause.

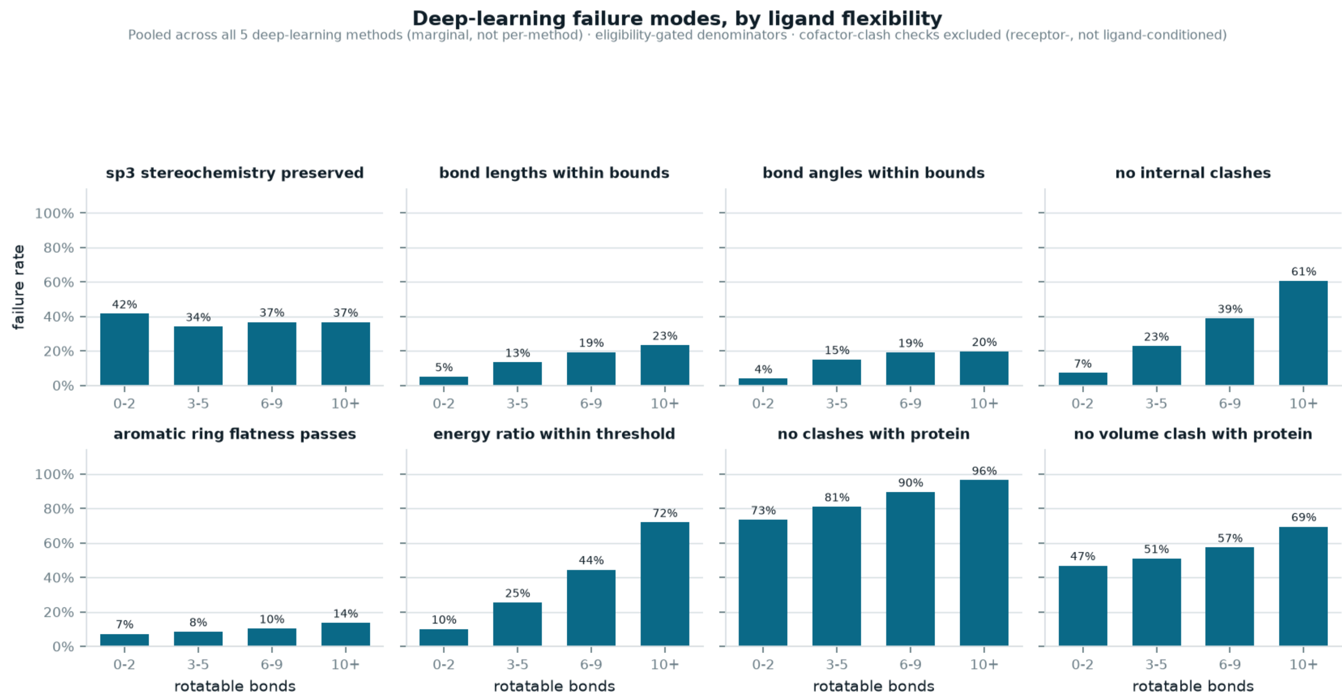

FIG 3 Per-check failure with eligibility-gated denominators — a ligand that cannot fail a check is excluded rather than counted as a trivial pass. Gating flattens sp3 stereochemistry almost completely; ungated it appeared to trend. The four cofactor checks are excluded: they are conditioned on the receptor, not the ligand.

Both would silently corrupt a naive re-analysis

Blank is not False. Both flatness columns are empty for all 2,996 energy-minimised rows — those checks were not run. Reading blanks as failures makes every minimised pose invalid, producing the nonsensical result that minimisation destroys 100% of poses including AutoDock Vina's. Handled correctly, minimisation takes DiffDock from `24.1% → 65.0%` valid while costing Vina `14.5 points`.

No output is not perfect output. 85 rows have every check blank — the method produced no pose at all. Since nothing is False, a naive "all checks passed" test scores them valid and inflates every validity rate.

What this data cannot answer

Descriptors describe the molecule that was docked, not the pose that came back — nothing here measures how *wrong* a pose is beyond the binary checks. 85 of 126 strata could not be estimated at all, so the 95 surviving effects describe roughly a third of the grid. The size-versus-flexibility contrast is not identified for DiffDock, and Gold and Vina fail too rarely to fit a model on most checks — these are not gaps more analysis of this dataset would close.

The predicted poses themselves were never released, so the methods' checks cannot be independently recomputed; only the crystal-structure rows could be, and were. The crystal-contact flag is this project's own estimate, bracketing but not reproducing the paper's undisclosed subset. Several surviving effects rest on small failure counts even after the ≥15-failure gate — check `n_fail` before treating any single row as decisive.

Data. PoseBusters paper data, Zenodo 8278563. Method: Buttenschoen M, Morris GM, Deane CM, *PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences*, Chem Sci 15, 3130 (2024), doi:10.1039/D3SC04185A.

Reproduce. Every table in this document is read from `reports/tables/` at render time; every figure is inlined from `reports/figures/`. Regenerate with `python -m pb.acquire` → `descriptors` → `crystal_contacts` → `build` → `validate` → `analyze` → `replication` → `figures` → `report_html`. 37 tests, 20 commits.